

$$\begin{bmatrix} u \\ v \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}}_{\text{Rigid body rotation}} \cdot \left(\underbrace{\begin{bmatrix} k_x - 1 & \gamma_x \\ \gamma_y & k_y - 1 \end{bmatrix}}_{\text{Uniform stretch and shear}} \cdot \begin{bmatrix} x \\ y \end{bmatrix} + \underbrace{\begin{bmatrix} u_x^g \\ u_y^g \end{bmatrix}}_{\text{2D Gaussian deformation}} \right) + \underbrace{\begin{bmatrix} t_x \\ t_y \end{bmatrix}}_{\text{Rigid body translation}} \quad (1)$$

$$\begin{bmatrix} u_x^g \\ u_y^g \end{bmatrix} = \sum_{j=1}^N \begin{bmatrix} A_x^j e^{-\frac{1}{2} \left(\frac{x-x_0^j}{\sigma_{x0}^j} \right)^2 - \frac{1}{2} \left(\frac{y-y_0^j}{\sigma_{y0}^j} \right)^2} \\ A_y^j e^{-\frac{1}{2} \left(\frac{y-y_1^j}{\sigma_{y1}^j} \right)^2 - \frac{1}{2} \left(\frac{x-x_1^j}{\sigma_{x1}^j} \right)^2} \end{bmatrix}, \quad N = 1 \text{ or } 2 \quad (2)$$